

CUTPURSE: Explanation-Guided Model Extraction on Tabular Data Survives Withholding the Explanation

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Abstract

Post-hoc explanations can improve model extraction, but it is unclear whether that advantage comes from information the explanation uniquely discloses or from information an adversary could obtain independently. We study this question in explanation-guided extraction of tabular classifiers. We decompose what an explanation contributes into an *attribution* channel, which identifies influential features, and a *threshold* channel, which supplies candidate coordinates at which the prediction may change, and ablate each while holding the rest of the attack fixed over ten independent target instantiations per comparison. Across six datasets and five target families, attribution provides no systematic extraction advantage, whereas LIME’s discretization thresholds can improve fidelity by up to 41 percentage points on continuous features. This finding motivates withholding the bin edges as a targeted defense. However, LIME’s default thresholds are quantiles of its background data rather than properties of the target model. An adversary with representative auxiliary data can therefore reconstruct an equivalent discretization without querying the explanation endpoint.

We use this observation to build CUTPURSE, an explanation-free extraction attack that reconstructs the discretization grid and combines it with diverse query generation and boundary search. Against the defended service, CUTPURSE improves fidelity by 16.3 percentage points on targets with continuous features and by 7.0 percentage points over the undefended published attack. It provides no advantage on purely categorical targets, where there is no continuous axis to discretize. Given an adversary with representative auxiliary data, withholding explanations therefore removes a convenience for extraction without removing the underlying extraction capability.

Keywords

model extraction, model stealing, explainable AI, SHAP, LIME, machine-learning-as-a-service, privacy

1 Introduction

Machine-learning services increasingly return a post-hoc *explanation* alongside each prediction. Cloud providers increasingly offer attribution APIs, while regulatory pressure, including the GDPR’s provisions on automated decision-making and the EU AI Act’s transparency obligations, further encourages their adoption. The premise is that a prediction accompanied by a reason is more accountable than a prediction alone.

However, explanations may also introduce a security risk because they provide additional information about the model an adversary wants to steal. Model extraction reconstructs a functional copy of a black-box target from query access [4, 19]. Such a surrogate can also enable further attacks, including membership inference, at a lower query cost [13].

Prior explanation-guided extraction attacks show that querying with explanations can produce higher-fidelity surrogates with fewer queries [1, 9, 12]. These results suggest that explanations help but they are typically reported end-to-end: the attack is run with the explanation and compared against a baseline without it. The resulting gain is therefore attributed to the model-specific information contained in the explanation, such as locally important features or, for gradient-based explanations, the local geometry of the decision boundary. However, an end-to-end comparison cannot distinguish this model-specific information from other information supplied by the explanation, such as convenient coordinates for placing the next query. Identifying the source of the gain matters because it determines whether the security risk comes from explanations themselves, from a particular explanation method, or from information an adversary could obtain independently.

We take Autolycus [12], a state-of-the-art explanation-guided extraction attack against interpretable models on tabular data, as our starting point and ask where its advantage actually comes from. We identify three weaknesses in this approach (§4). *First*, its traversal explores only outward from the auxiliary seeds it starts with, so whole regions of the target’s decision space are never queried. As a result, how much of the model it recovers depends on the seed set rather than solely on the query budget, even though query diversity is an established strategy for improving model extraction [16]. *Second*, the traversal takes no active steps to locate decision boundaries, even though selecting queries for proximity to the boundary is a well-established extraction principle [16, 19]. *Third*, the model-specific information in the explanation does not appear to drive the extraction gain: choosing features to perturb based on SHAP importance is indistinguishable from choosing them at random, and LIME’s ranking behaves the same way. Instead, the gain appears to come from a narrower source: the *discretization threshold* that LIME exposes as a bin edge, which indicates where along a feature the prediction may change.

A natural defense, and why it fails. If the threshold is the source of the extraction gain, a natural mitigation is to withhold it: the service could return the attribution weights while suppressing the bin edges. This reduces the performance of the attack it targets: Autolycus loses up to 41 fidelity points (§6.3). However, this defense does not prevent an adversary from reconstructing the same thresholds. LIME’s default discretizer places its cuts at the 25th, 50th and 75th percentiles of the background data. These thresholds are properties of the data distribution, not of the model. Given a sufficiently representative auxiliary sample, an adversary can estimate

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an equivalent grid itself and recover the withheld gain, leaving the target’s own discretizer no measurable advantage (§6.4, which also bounds how small that sample can be before the estimate degrades). The mitigation withholds a statistic the adversary can estimate for itself.

We therefore introduce CUTPURSE, an extraction attack that combines a self computed discretization grid with the two capabilities Autolycus lacks: diverse query generation to improve the coverage and boundary search to improve query placement. CUTPURSE queries no explanation endpoint at all. Against a service running the threshold defense, CUTPURSE improves the fidelity by a mean of +16.3 percentage points on targets with continuous features. Against the *undefended* service, where Autolycus receives the full explanation, it still improves the fidelity by +7.0 percentage points on average. Thus, the defense reduces the transparency of the service without reducing the adversary’s extraction performance. Continuous features are where the threshold channel exists at all, and so where both the leak and its mitigation are meaningful (§7).

Contributions.

- An analysis of the state-of-the-art explanation-guided extraction attack that traces its advantage to a single channel. The *attribution* channel, which identifies the features considered important by the explanation, provides no systematic advantage for either SHAP or LIME. When explanation-guided feature choice has a significant effect, the direction of that effect depends on the target rather than on the explanation method (§6.2). The threshold channel provides the extraction gain, improving fidelity by up to 41 percentage points (§6.3).
- An evaluation of the mitigation that withholds LIME’s bin edges, and a demonstration that it fails: the edges are quantiles of the background distribution, so an adversary can reconstruct an equivalent grid from an auxiliary sample without measurable loss (§6.4). Across both the grid-level and attack-level comparisons, the adversary’s reconstructed grid performs indistinguishably from the target’s grid. These controlled comparisons, rather than comparisons between different attacks, test whether the explanation discloses information that the adversary could not obtain independently.
- CUTPURSE, an extraction attack that queries no explanation endpoint and combines a self-computed grid with diverse query generation and boundary search. Its components are established extraction techniques. Our contribution is to show which components are needed to replace the information provided by the explanation, and that the required coordinates can be computed independently. It improves the fidelity by +16.3pp over threshold-defended Autolycus and by +7.0pp over undefended Autolycus on continuous-feature targets. These results concern attack strength rather than disclosure and it can also target prediction-only services, which explanation-guided attacks cannot target.
- An evaluation framework separating *what an adversary needs from what a service provides it*: The framework compares four adversaries that differ only in whether they have the target’s explanation, their own grid, or neither. All four

are evaluated under the same query budget, seed set, and evaluation set, with ten independent target instantiations per cell (§6.5).

Scope. We study one family of explanation-guided extraction attacks against interpretable classifiers [12] on tabular data, using two explainers in their default configurations. We therefore claim that post-hoc explanations disclose less information than the literature assumes *in this setting*, not that they are harmless in general. A differently designed attack, or a discretizer fitted on labels rather than on the marginal distribution, could plausibly extract more from the same explanations, and we do not establish that our results transfer to arbitrary explainers or discretizer configurations, to high-dimensional or neural targets, or to an adversary without representative auxiliary data.

2 Background and Related Work

2.1 Model Extraction and Post-hoc Explanations

Model extraction attacks reconstruct a functional copy $\hat{f}$ of a black-box target f from query access alone, with success measured by *prediction agreement* between $\hat{f}$ and f on held-out data (formalised in §3). When the target is deployed as an explainable service, each query returns not only a label but a post-hoc *explanation*, which may reveal additional structure about f . Two dominant explanation families expose fundamentally different information:

- **Attribution methods (SHAP).** SHAP [8] assigns each feature a signed magnitude $\phi_i(x)$, its Shapley marginal contribution to the prediction at x . This is a statement of *how much* each feature matters locally. It carries no explicit information about *where* along a feature axis the decision changes.
- **Local surrogate methods (LIME).** LIME fits an interpretable model to a *discretized* neighbourhood of x : continuous features are binned, and the returned coefficients are attached to (feature, bin) pairs. LIME therefore reveals both an influential feature *and* the bin edge, a *threshold*, that separates neighbouring regions.

This distinction between *magnitude-only* and *threshold-bearing* explanations is central to our analysis: as we show, it is the threshold, not the attribution magnitude, that an extraction attacker can exploit (§3, §4).

2.2 Related Work

Model Extraction Attacks. Model extraction attacks aim to reconstruct a functional copy of a black-box target model through strategic querying [19]. Early work by Tramèr et al. [19] demonstrated that simple equation-solving over prediction outputs suffices for linear models. Subsequent methods scaled to neural networks: Knockoff Nets [14] trains a surrogate on outputs obtained by querying with a knock-off dataset, while Jagielski et al. [4] show that high-fidelity extraction of deep networks is possible given sufficient queries. A parallel line of work [2, 17] frames model extraction as knowledge distillation, transferring decision boundaries through synthetic queries. Marich [5] formalises the problem as *distributional equivalence* and proposes an active sampling strategy that maximises information gain per query. In contrast, our method

requires only a small per-class seed set rather than a large public transfer set, and synthesises informative queries from that seed.

Explanation-Guided Extraction. The observation that post-hoc explanations leak structural information about the target model motivates a distinct attack family. Milli et al. [9] first demonstrated that gradient-based explanations suffice to reconstruct linear models exactly. Miura et al. [10] (MEGEX) carry gradient explanations into the *data-free* setting: treating the returned gradient as a derivative signal, the adversary trains a generative model that synthesises its own queries, so no real input data is needed. Both exploit a property our SHAP and LIME setting lacks, namely that a gradient specifies a *direction* in input space rather than only a per-feature magnitude. Autolytus [12] extended this to classical ML classifiers (decision trees, logistic regression, naive Bayes, and k -NN) by using LIME and SHAP explanations as traversal guides: the attacker perturbs seed samples along the directions of highest feature attribution, progressively labelling a synthetic dataset to train a surrogate. Yan et al. [21] (XaMEA) similarly exploit the spatial information contained in explanations in a model-agnostic manner. A common assumption, implicit in these attacks, is that the explanation’s *attribution ranking* is what drives the extraction gain. Through controlled ablations we revisit this assumption and find it does *not* hold: the gain stems primarily from boundary-directed query synthesis (boundary search and diverse sampling), while the choice of features by attribution contributes little beyond a random choice. The explanation supplies additional signal only when it exposes a discretization *threshold*: LIME’s discretization surfaces such thresholds directly, whereas SHAP’s magnitude attribution does not, so a SHAP-guided attacker must *search* for the boundary and gains little over the sampling procedure itself. This reframes explanation leakage as a matter of decision *thresholds* rather than boundary *geometry*.

Counterfactual-Based Extraction. A growing body of work exploits counterfactual explanations (CFs) rather than attribution scores. Aivodji et al. [1] first showed that CFs leak enough boundary information to build high-fidelity surrogates, and DualCF [20] improved query efficiency by training on CF/counter-CF *pairs* that straddle the decision boundary. Khouna et al. [6] propose the Tree Reconstruction Attack (TRA), which uses a *restricted* CF oracle to perform axis-aligned binary search, achieving exact functional reconstruction of decision trees and random forests. Although theoretically elegant, TRA requires an oracle that returns the *nearest* counterfactual confined to a specific hyper-rectangle, an assumption rarely satisfied by deployed XAI APIs. Concurrently, Otto et al. [15] derive query complexity bounds for extracting linear classifiers under factual and counterfactual query regimes. Our work connects these lines: rather than a bespoke CF oracle, we obtain boundary thresholds from the explanation itself (directly from LIME’s discretization, or by SHAP-directed single-axis bisection) without requiring restricted oracle access. Consistent with our central finding, it is this threshold signal, not the attribution magnitude, that constitutes the exploitable channel.

Privacy Implications of Explanations. Recent work highlights the security cost of XAI transparency. Ezzeddine et al. [3] show that exposing counterfactual explanations through query-based APIs

makes shadow-based membership inference attacks (MIAs) more effective in MLaaS settings, and characterise a three-way trade-off between privacy leakage, predictive performance and explanation quality. Quan et al. [18] make the complementary point that post-hoc feature-importance explanations act as a side channel that lowers the query cost of evasion, membership inference and extraction alike. LoMime [13], from the authors of Autolytus, uses surrogate model extraction as a stepping stone for label-only MIA, demonstrating that functionally equivalent surrogates enable downstream privacy attacks with far fewer queries than direct methods. Our extraction fidelity thus serves as an upstream component for such attacks. Surveys of privacy-preserving explanation [11] catalogue countermeasures. Localising the effect to a specific channel (the discretization threshold) lets us ask what such a countermeasure would have to accomplish. We find that obfuscating that channel is unlikely to suffice, since the adversary can reconstruct an equivalent grid from its own data (§6.4).

3 Problem Formulation

Setting. Let $f : \mathcal{X} \rightarrow \mathcal{Y}$ be a black-box target classifier, with $\mathcal{X} \subseteq \mathbb{R}^d$ and $\mathcal{Y} = \{0, \dots, C-1\}$. The attacker knows neither the architecture of f nor its training data, and interacts with f only through an explanation-augmented prediction API. Each query $x \in \mathcal{X}$ returns (i) the predicted label $f(x)$ and the class-probability vector $\hat{p}(x) \in \Delta^C$, and (ii) a post-hoc explanation $e(x)$ produced by a fixed explainer $\mathcal{E} \in \{\text{SHAP}, \text{LIME}\}$. Every call counts against a total query budget Q . The attacker additionally holds a small per-class *seed set* $\mathcal{D}_0 = \{(x_i, y_i)\}$ with n examples per class (following the respective Autolytus conventions, $n=1$ for LIME and $n=5$ for SHAP), drawn from a public split that shares $\mathcal{X}$ ’s feature schema. The attacker does not have access to a large public transfer set

Security property and privacy implication. The property under attack is *model confidentiality*: the adversary’s goal is a surrogate that agrees with f , and agreement on held-out data is what we measure throughout. The privacy consequence is downstream and indirect. A higher-fidelity surrogate is a better basis for membership and attribute inference against the target’s training data [11, 13], which is what makes extraction a privacy question and not only an intellectual-property one. We inherit that link from prior work rather than demonstrating it: no experiment in this paper measures a downstream inference attack. Accordingly, claims about privacy in this paper refer to model confidentiality and its established downstream connection to inference attacks

Explanation oracle. The two explainers expose structurally different information about f near x :

- **SHAP:** $e(x) = \phi(x) \in \mathbb{R}^d$, a vector of signed Shapley attributions in which $|\phi_i(x)|$ measures the local influence *magnitude* of feature i [8]. It reports *how much* each feature matters, but not *where* the decision changes along it.
- **LIME:** $e(x) = \{(i, [l_i, u_i], w_i)\}_{i \in F}$ over a selected feature subset F , where feature i has local weight w_i and lies in the discretization bin $[l_i, u_i]$. Beyond the feature weight, LIME therefore exposes a *bin edge* that provides a candidate coordinate for locating changes in f ’s output.

Objective. The attacker trains a surrogate $\hat{f} : \mathcal{X} \rightarrow \mathcal{Y}$ that is functionally equivalent to f , measured by prediction agreement:

$$\text{Sim}(\hat{f}, f) = \mathbb{E}_{x \sim P_X} [\mathbf{1}(\hat{f}(x) = f(x))]. \quad (1)$$

Here P_X is the (unknown) data distribution over $\mathcal{X}$, estimated in practice by a held-out test set disjoint from the attacker’s seed set $\mathcal{D}_0$ and its queries. The attack succeeds if $\text{Sim}(\hat{f}, f)$ is high while the number of API calls stays within Q .

What does an explanation leak? The effectiveness of an extraction attack rests on *query placement*: labelled queries concentrated at the decision boundary and spread across diverse decision regions can provide more information for training a surrogate than queries drawn from the interior of a decision region, independent of any explanation [16, 19]. Our question is, therefore, what a post-hoc explanation adds to this placement. We decompose the explanation’s possible contribution into two channels:

- (1) **Attribution:** which features are locally influential (a ranking by $|\phi_i|$ or $|w_i|$), available from both SHAP and LIME.
- (2) **Threshold:** where along a feature axis the prediction changes, i.e. the coordinate of the decision boundary. LIME’s bin edge $\{l_i, u_i\}$ is a direct estimate, while SHAP’s magnitude carries no such location.

We formulate a separate hypothesis for each channel. Our experiments find no systematic advantage from the attribution channel (§6.2), while the threshold channel is the one that provides the observed advantage (§6.3), but is reproducible from the adversary’s own data rather than being uniquely disclosed by the service (§6.4). We treat that pair of findings as a design specification rather than as a conclusion, and build the attack it implies (§5, §6.5).

Neither channel is an attack in itself. The query-placement machinery of §5 (boundary search and diverse generation) is a standard extraction technique that the explanation does not disclose and that an attacker can run without any explanation at all. What an explanation can change is only *where* that machinery directs its queries: the attribution channel proposes which coordinate to move along, and the threshold channel proposes how far to move it. Isolating these two inputs is therefore the natural way to ask what the explanation itself contributes. Prior explanation-guided attacks [9, 12] implicitly credit their gain to the *attribution* channel and, for gradient explanations, to recovering boundary *geometry* (the local normal). Our central hypotheses, examined in §4 and §6, are:

H1 (attribution is not systematically advantageous). Steering

query placement by explanation importance does not improve fidelity over a random feature choice, on average or reliably in sign. Particular target families may be exceptions, and §6.2 identifies one; the claim is that the adversary gains nothing it can count on in advance, not that the channel is empty.

H2 (the threshold is the operative channel). The only explanation supplied signal that improves fidelity is the threshold channel. A threshold-bearing explainer (LIME) helps, whereas a magnitude-only explainer (SHAP) does not, because, lacking a threshold, it must *search* for the boundary.

H3 (the threshold is not a disclosure). The threshold that helps is a statistic of the *data distribution* (LIME’s default bin

edges are quantiles of its background sample, independent of the model and of the labels) rather than a property of the target. An adversary holding a modest auxiliary sample can therefore reconstruct an equivalent grid, so the channel accelerates extraction without the service disclosing anything the adversary could not compute itself.

H1 and H2 together locate *which* part of an explanation an attacker consumes. H3 asks whether consuming it required the explanation at all. That last question is the organising principle of this paper: leakage should be measured against what the adversary could obtain unaided, not against nothing.

Threat model. We assume: (1) black-box access only (no white-box gradients); (2) the API returns post-hoc explanations (SHAP or LIME), as increasingly expected for transparency (e.g. GDPR Article 22); (3) the attacker knows the feature schema but not the model family or hyperparameters; and (4) the attacker holds an auxiliary sample D_A drawn from the target’s distribution, from which it draws $n \leq 5$ seeds per class. Assumption (4) is inherited unchanged from the attack we analyse [12], which allows D_A to range “from a rudimentary collection of samples to comprehensive datasets”. §6.4 shows the attack needs only the rudimentary end, so nothing below turns on the generous reading of that range. D_A has exactly two roles, supplying the seed set and fitting the adversary’s discretizer. Seeds enter as ordinary queries, labelled by the target rather than by their own ground truth, and the remainder of D_A is never queried, never labelled and never used to train the surrogate, which sees only (query, prediction) pairs.

4 Three Weaknesses of Explanation-Guided Traversal

Autolycus expands a small per-class auxiliary seed set by repeatedly explaining the current sample, perturbing its top-ranked features, and querying the results. This section sets out three limitations of that design. The first two are limitations of the *traversal*, and motivate the two components CUTPURSE adds (§5). The third concerns the *explanation* itself, and motivates the rest of the paper. §6 tests all three.

W1: coverage is a property of the seed set, not the budget. The traversal only ever moves outward from where it started: every queried point is a perturbation of a previously queried point, seeded from the auxiliary sample. Regions of the input space that no seed lies near are therefore never reached, however large the budget. Surrogate errors concentrate accordingly, not only near the boundary, where they are expected, but in class *interiors* the traversal never visited, where the target is confident and the surrogate is confidently wrong. A budget spent extending existing trajectories cannot fix this. Coverage has to be created explicitly, which is what Phase 1 does.

W2: the traversal never searches for a boundary. Extraction is most query-efficient when labelled points sit *on* the decision surface: Tramèr et al.’s line-search retraining selects points closest to the current surrogate’s boundary, and ACTIVETHIEF reaches the same regions through uncertainty [16, 19]. Points straddling a boundary constrain the surrogate far more than points drawn from a class interior. Autolycus places queries by stepping $\pm \epsilon$ from the

current sample along explanation-selected features, which crosses a boundary only incidentally. No part of the attack takes a pair of oppositely-classified points and searches between them, even though doing so costs a handful of queries and yields a point arbitrarily close to the surface. Phase 2 supplies that search.

W3: attribution is magnitude, not location. An attribution vector $\phi(x)$ ranks features by local influence but says nothing about *where* along a feature the prediction changes. Two attackers handed the same ranking but stepping to different coordinates place entirely different queries, so the ranking alone cannot determine query placement. Empirically the ranking is not merely insufficient but unreliable: replacing the explanation’s top- k choice with a random one leaves surrogate fidelity centred on zero, for SHAP and LIME alike, with the target-specific exceptions of §6.2.

What an attacker *can* exploit is a *threshold*: a candidate coordinate to move a feature to. LIME’s discretization supplies one for free (each returned (feature, bin) pair carries the edges of the bin the current value falls in), so a LIME attacker can place queries at candidate cut points without spending any to find them, while SHAP, carrying only magnitudes, must *search* for a crossing and spend queries recovering what LIME supplies for nothing. This predicts that destroying the threshold collapses the advantage, which §6.3 confirms. Explanation leakage in this setting is thus a matter of discretization *thresholds*, not of attribution.

The mitigation W3 implies, and its flaw. W3 suggests a targeted defense: return the attribution weights but withhold the bin edges. It is effective against Autolytus (§6.3). Its flaw is that a bin edge is not a model secret. LIME’s default `QuartileDiscretizer` places its cuts at the 25th, 50th and 75th percentiles of the background data it was fitted on, a statistic of the data distribution rather than of the model or its labels, and one that can be estimated from any sufficiently representative sample. An adversary granted auxiliary data by the threat model can therefore estimate an equivalent grid and recover the withheld gain, subject to the sample-size bound of §6.4. CUTPURSE is the attack that does so, together with the two components W1 and W2 call for.

5 CUTPURSE

CUTPURSE answers the three weaknesses of §4 with three phases, and takes its coordinates from a grid it computes itself rather than from the service. *Phase 1* (diverse generation) creates coverage the traversal cannot reach on its own, addressing W1. *Phase 2* (boundary search) takes oppositely-classified pairs and bisects between them, addressing W2. *Phase 3* is the traversal itself. Phases 2 and 3 need a coordinate to move a feature *to*. Where Autolytus reads that coordinate from LIME’s returned bin edge, CUTPURSE reads it from a `QuartileDiscretizer` fitted on its own auxiliary sample, which per W3 is an equivalent statistic. No phase queries an explanation endpoint: Phase 1 takes neither a model nor an explainer, and Phases 2 and 3 consume only bin edges, which the adversary reads from its own discretizer at construction time. Diverse query synthesis and boundary-directed bisection are established extraction techniques rather than novel ones [16, 17, 19]; what this section contributes is not the components but the consequence of §4, that

Algorithm 1 Boundary-directed extraction with explainer $\mathcal{E} \in \{\text{SHAP}, \text{LIME}\}$. Parameter values are given in §5.1–5.3.

Require: target f , explainer $\mathcal{E}$, seeds $\mathcal{D}_0$, budget Q

Ensure: surrogate $\hat{f}$

```

1:  $\mathcal{S}, \mathcal{T} \leftarrow \mathcal{D}_0$  ▷ traversal queue, labelled set
   // Phase 1: diverse query generation
2:  $\mathcal{S} \leftarrow \mathcal{S} \cup \{x \in \text{DIVERSEGEN}(\mathcal{S}) : \max \hat{p}(x) > \tau\}$  ▷  $\mathcal{E}$  used
   only by the LIME-threshold variant
   // Phase 2: boundary search
3: for  $n_{\text{mid}}$  cross-class pairs  $(a, b)$  drawn from  $\mathcal{S}$  do
4:    $c \leftarrow a$  with its top- $k_b$  features moved toward  $b$  ▷ LIME: to
   the bin edge; SHAP: midpoint
5:    $(c, M) \leftarrow \text{BISECT}(c, b)$  ▷ 8 steps, midpoints  $M$ 
6:    $\mathcal{S} \leftarrow \mathcal{S} \cup \{c\}; \mathcal{T} \leftarrow \mathcal{T} \cup M$ 
7: end for
   // Phase 3: explanation-guided traversal
8: while  $\mathcal{S} \neq \emptyset$  and budget and per-class quotas remain do
9:    $x \leftarrow \text{POP}(\mathcal{S}); \mathcal{T} \leftarrow \mathcal{T} \cup \{(x, f(x))\}$ 
10:  for all  $i$  among the top- $k$  features of  $x$  by  $\mathcal{E}$  do
11:    add  $x$  with coordinate  $i$  stepped by  $\pm \varepsilon_i$ , if valid and new
    ▷ LIME: from the bin edge
12:  end for
13: end while
14: return  $\hat{f} \leftarrow \text{TRAIN}(\mathcal{T})$ 
```

once the threshold is recognised as a self-computable statistic, no explanation endpoint is needed to drive them.

Figure 1 sketches the pipeline. CUTPURSE builds on the Autolytus traversal, which expands a small per-class seed set one feature at a time: it picks a *base coordinate* for a selected feature and emits two children, one at each of $\text{base} \pm \varepsilon_i$, where ε_i is a fixed per-feature step scaled to feature i ’s spread. Which coordinate serves as the base is precisely what the explainer supplies: LIME offers a bin edge and SHAP offers nothing, leaving the feature’s current value (§5.3). Like a bin edge, ε_i is a distributional statistic, so the adversary can set it from its own auxiliary sample.

The three phases can be driven with either explainer $\mathcal{E} \in \{\text{SHAP}, \text{LIME}\}$ and the two instantiations differ *only* where the discretization threshold enters the computation (Phase 2 placement and Phase 3 stepping). The bin-edge stepping that constitutes LIME’s threshold channel is Autolytus’s *own* mechanism, retained unchanged rather than introduced by us. Holding everything else fixed turns the SHAP-versus-LIME comparison into a controlled experiment that isolates the threshold channel (**H2**), while ablating the explanation’s feature choice isolates the attribution channel (**H1**).

A variant reading the target’s LIME output instead of a self-computed grid lets query *placement* be compared against Autolytus at equal information access (§B, Appendix A). The channel ablations are separate again: they run on the *base* Autolytus traversal, so those conclusions do not depend on our additions. Algorithm 1 summarises the three phases.

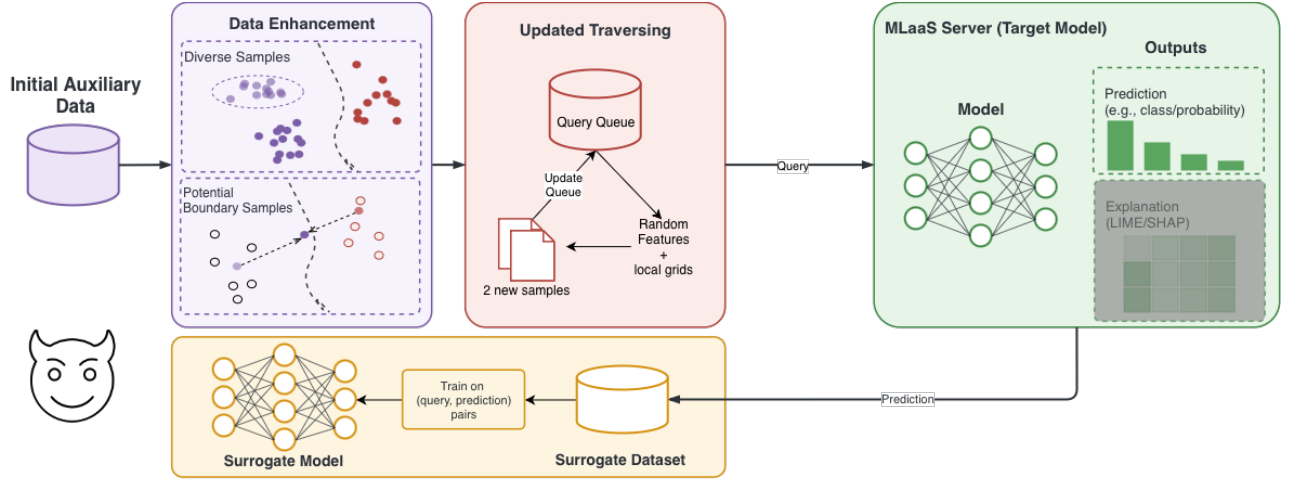


Figure 1: CUTPURSE. Phases 1 and 2 add diverse and boundary samples to a small seed set; Phase 3 steps randomly chosen features from bin edges the adversary reads off a grid fitted on its own auxiliary data. The explanation endpoint (greyed) is never queried. The ablations of §6.2–§6.3 read it instead.

5.1 Phase 1: Diverse Query Generation

Autolycus seeds its traversal from a handful of per-class points and expands them by local perturbation, which tends to under-cover behavioural regions the seeds never reach. Phase 1 counters this with one of two generators, both returning *unqueried* candidates that are then queried and kept only if confidently classified ($\max \hat{p}(x) > \tau$ with $\tau = 0.8$):

- **Plausibility-filtered** (explanation-free): crossover and mutation of the current pool produce a large candidate set, of which those further from the seeds than the 90th percentile of the seeds’ own nearest-neighbour distances are rejected as off-manifold. The survivor furthest from everything kept so far is then added to the pool, and the step repeats. This is a greedy max–min (k-center) selection in input space, purely geometric and using no model output beyond the confidence filter above.
- **LIME-threshold** (explanation-informed): harvest the target’s LIME bin-edge thresholds from the seed explanations, synthesise seed-anchored candidates that *cross* those thresholds into different decision cells, plausibility-filter them, and greedily select the candidates covering the most distinct (feature, threshold, side) configurations. If the target surfaces no thresholds (e.g. a linear model), it falls back to the geometric generator.

In its plausibility-filtered form Phase 1 uses no explanation information at all, and this is the form CUTPURSE runs. The LIME-threshold variant uses both channels: the ranking decides whose thresholds are harvested (only the top- k features of each seed explanation are read), and the thresholds themselves drive the crossing.

5.2 Phase 2: Boundary Search

Phase 2 places training points *on* the decision boundary, where labels are most informative. We enumerate the cross-class pairs available among the current samples, shuffle them, and draw n_{mid}

of them without replacement (refilling the pool if it empties). For each drawn pair (a, b) we build a child from a by blending its top- k_b features toward b , then run an 8-step bisection between the child and b to converge onto the boundary between their classes. Enumerating every pair would be quadratic in the seed count and would exhaust the budget on this phase alone, so n_{mid} is a parameter (10 by default, reduced at small Q so Phase 3 is not starved). The explainer governs *how* the child is placed, and this is precisely where the threshold channel enters:

- **LIME** snaps each continuous top- k_b feature to its local *bin edge* on the side toward b , a direct, free estimate of where the boundary crosses that coordinate.
- **SHAP**, having only magnitudes, places the blend at the *midpoint* and must rely on the subsequent bisection to *search* for the same boundary location.

The bisection midpoints are recycled into $\mathcal{T}$ as boundary-concentrated training data, and are not ϵ -perturbed further.

5.3 Phase 3: Explanation-Guided Traversal

Phase 3 is the Autolycus-style expansion. We pop a sample x , query and label it with $\mathcal{E}$, rank its features by $|\phi_i|$ (SHAP) or $|w_i|$ (LIME) and keep the top k , then emit two children per selected feature i , stepped by $\pm \epsilon_i$. The attributions choose *which* coordinates move; ϵ_i sets how far. Valid, non-duplicate children re-enter the queue until the budget is exhausted or all classes are sufficiently visited. We retain this expansion because it matters on its own (removing Phase 3 degrades fidelity, most sharply for tree targets), but the *attribution-based selection* is *not* what drives it: consistent with **H1**, replacing the explanation’s top- k choice with a random one leaves fidelity essentially unchanged on our primary (non-tree) targets (§6.2). What differentiates the two explainers here is, once more, the threshold channel: LIME steps $\pm \epsilon_i$ *from the feature’s bin edge*, so each step crosses the local threshold, whereas SHAP steps from the current value using only the attribution ranking. Phase 3

therefore contributes coverage and expansion. Its sole explanation-supplied benefit is LIME’s threshold stepping, not the attribution ranking.

6 Experiments

6.1 Setup

We evaluate on six public tabular datasets (Table 1) covering continuous, categorical, and mixed feature schemas, against five target families: logistic regression (LR), naive Bayes (NB), k -NN, decision tree (DT) and random forest (RF). All five appear in every experiment: the channel ablations of §6.2 and §6.3, the self-computability test of §6.4, in the adversary comparison of §6.5 and in the budget sweep. We scope to tabular data throughout, following the attack we analyse [12]. Following the Autolytus convention, the LIME attacker uses $n=1$ seed per class and explores $k=3$ top features, and the SHAP attacker uses $n=5$ and $k=3$. Query budgets are per dataset (Table 1). We inherit the baseline’s query accounting unchanged: the explanation is bundled with the prediction and not charged separately, a candidate already queried or queued is not resubmitted, and fidelity is reported against the query *budget* rather than queries consumed. Phases 1 and 2 both consume budget: every diverse candidate is queried, including those the confidence filter then discards, and every bisection midpoint is queried and retained as training data. A budget-scaling rule caps that combined overhead so it cannot starve Phase 3 at small Q . Every arm we compare spends the same counted budget, so CUTPURSE claims no advantage from the accounting.

Compared arms always run on *paired* seed sets (identical draws, identical RNG state), and surrogate fidelity (Eq. 1) is aggregated by the *mean* over refits. Significance is a paired Wilcoxon signed-rank test over the 10 paired observations, with $*p<.05$ and $**p<.01$. With 10 pairs the smallest attainable two-sided p -value is $2/2^{10} \approx .002$, reached only when every pair falls the same way, so $**$ marks a near-unanimous effect across target instantiations rather than an arbitrarily small p .

Protocol. We deviate from the original attack’s reporting in two ways, both toward more conservative estimates. *Surrogate aggregation:* the original trains R surrogate refits per arm and reports the single one most similar to the target, a quantity measured on the evaluation set, so the selection is made using the reported metric. We instead report the mean over the R refits, which is unbiased. (R denotes surrogate refits and is unrelated to the top- k feature budget k , and we use $R=20$ for DT, $R=10$ for RF, and $R=1$ otherwise.) *Target instantiation:* the original varies only the attacker’s seed sets against a single fitted target, so its reported spread reflects seed noise rather than variation across targets. For the ablations (§6.2, §6.3) we therefore draw 10 independent train/test splits and retrain the target on each, pairing all arms within a split, so each observation is an independent target instantiation. We retain the original single-split protocol only for the decomposition of Appendix A and the sweeps of Appendix B and Appendix D, so that those numbers stay comparable to the published baseline. Every comparison in the main text uses the multi-split protocol.

Table 1: Datasets. *cont.* counts continuous features and Q is the query budget. Train, evaluation and auxiliary partitions follow the 75:15:10 split of the attack we analyse.

	n	feat.	cont.	cls.	Q
crop	1,700	7	7	17	1000
adult	32,561	11	3	2	1000
breast	569	30	30	2	100
nursery	12,960	8	0	3	1000
mushroom	8,123	21	0	2	1000
pendigits	10,992	16	16	10	1000

6.2 Is the attribution channel the leak? (H1)

Base Autolytus has two binary knobs: whether the top- k features come from the explanation’s ranking or at random, and whether steps start from the exposed bin edge or from the sample’s current value. This section varies the first, §6.3 the second. We replace the explanation’s top- k feature selection with a random- k choice, holding the rest of the attack fixed, and measure the change in surrogate fidelity. If the attribution ranking carries exploitable signal, random selection should be worse. Across both explainers the attribution channel is *weak and inconsistent* ($S=10$ paired sets):

- **SHAP: no net advantage.** On the base Autolytus attack, SHAP-guided selection is *no better on average* than random no-explanation traversal (Table 2: mean -0.2pp over 30 cells, with 4 significantly positive and 7 significantly negative). Where it helps it is confined to *tree* targets on datasets with a few dominant features, for which the *exact* TreeSHAP pinpoints true splits (MUSHROOM DT $+9.8$, RF $+6.9\text{pp}$, both $p<.01$). But the same explanation that helps trees on MUSHROOM *hurts* logistic regression on that very dataset (-11.2pp), and it misleads across the board on NURSERY (LR -10.2 , NB -9.1 , KNN -4.6pp). The sign of the effect is a property of the target, not of the explanation, so an attacker choosing to consult SHAP has no way to know whether doing so will help or harm.
- **LIME: no systematic advantage either.** Replacing LIME’s ranking with a random choice, holding the threshold available to both arms, changes fidelity by a mean of -0.6pp and a median of -1.0pp over the same 30 cells, with 4 significantly positive and 5 significantly negative (Table 3). This is weaker than claiming the channel is empty: individual cells show real effects, but their sign is not predictable from the dataset or the target family, so an attacker gains nothing it can rely on. The instability is itself informative: individual cells move by several points under changes that affect only *which* features the control happens to draw, which is what one expects when the quantity being measured is close to zero.

So neither explainer’s attribution ranking confers a systematic extraction advantage. This is not merely an absence of significance: the contrast with the *threshold* channel is *directional*. Over the identical cells, threshold ablation is significantly positive twelve times and significantly negative *never* (mean $+6.7\text{pp}$), whereas attribution is significant in both directions, four times positive against five

Table 2: Attribution ablation on the *base* Autolycus SHAP attack: fidelity gain (points) from selecting top- k features by SHAP importance rather than at random ($n=5$, $k=3$; $*p<.05$, $p<.01$). Ten paired splits per cell. Negative = random wins.**

	crop	pendigits	breast	adult	nursery	mushroom
LR	-5.5*	-1.9*	+1.2	-0.0	-10.2**	-11.2**
NB	-1.0	+0.6	+1.4	+2.5	-9.1**	+3.8
KNN	+1.0	+1.0	-0.3	+1.0	-4.6**	+3.6*
DT	+1.3	+2.7	+2.3	+0.7	-1.1*	+9.8**
RF	+0.4	-1.5	+0.3	+0.7	+0.7*	+6.9**

Table 3: Attribution ablation on the *base* Autolycus LIME attack: fidelity gain (points) from selecting top- k features by LIME’s ranking rather than at random, with the bin edge available to both arms ($n=1$, $k=3$; $*p<.05$, $p<.01$). Ten paired splits per cell. † marks cells with too few non-tied splits to test.**

	crop	pendigits	breast	adult	nursery	mushroom
LR	-3.8*	+0.1	+6.0	-4.5	-0.2	-2.9
NB	+6.8**	-1.4	+5.2	-1.8 †	-6.0	-15.4*
KNN	+4.4	-0.6	+4.0 †	-0.3	+5.7**	+6.1*
DT	-2.2	-5.7*	-2.7	-6.0	+1.5	+5.3
RF	-4.0	-6.9**	-5.6	-3.7**	+1.5*	+7.7

Table 4: Threshold ablation on the *base* Autolycus LIME attack: fidelity gain (points) from perturbing at the exposed *bin edge* rather than the current value, features held to LIME’s ranking ($n=1$, $k=3$; $*p<.05$, $p<.01$, † too few non-tied splits to test). Same protocol, cells and layout as Table 3. Positive = the threshold helps.**

	crop	pendigits	breast	adult	nursery	mushroom
LR	+40.9**	+18.6**	-10.2	-1.8	-0.8	-8.7
NB	+24.9**	+7.9**	+11.2*	+0.0 †	+5.0**	-0.2
KNN	+15.2**	+0.4	-0.2 †	-0.3	+1.0**	+13.9**
DT	+38.0**	+10.1*	+4.8 †	+2.6	+0.6	+2.1
RF	+22.7**	+0.7	+1.1	+0.9	+0.5	-0.8

negative, around a mean of -0.6 pp (§6.3). One channel has a sign, and the other does not. What LIME supplies is not a useful feature ranking but a discretization *threshold*.

6.3 Is the threshold (bin edge) the leak? (H2)

We now vary the second knob, with the feature choice fixed to LIME’s ranking: we perturb from the bin edge LIME exposes (default) versus from the sample’s current value (no-threshold), everything else identical. Because the base attack already snaps to the bin edge and has *no* boundary search to mask the effect, this is the cleanest possible test of H2.

The threshold channel is directionally consistent. On the two datasets whose evaluation sets can resolve it, the bin-edge signal produces large, significant gains: crop (agronomy, +15 to +41pp) and pendigits (pen-trajectory, +8 to +19pp), across ten independent target instantiations. The asymmetry is what matters: over these cells the threshold ablation is significantly positive twelve times and significantly negative *never*, whereas the attribution ablation on the identical cells is significant in both directions (Table 3). This is precisely the signal *absent* from SHAP: on pendigits, LIME’s threshold yields +8 to +19pp while SHAP’s attribution yields +2.7pp (n.s.), because Shapley magnitudes carry no “where” information. H2 holds where H1 fails.

Scope: the mechanism predicts where the channel is weak. The threshold channel is load-bearing only when LIME’s per-feature bin edge is a meaningful proxy for the decision boundary *and* the base attack has room to improve. Both conditions bite. Where features are *categorical*, discretization yields few distinct cut points and the channel is correspondingly weak and unstable: on NURSERY the effect is at most a few points against +15 to +41pp on crop, and it does not reproduce reliably (the NB cell measures +5.0 in one run and +1.0 in another differing only in random seed). We report this rather than a clean null because NURSERY’s features are label-encoded ordinals, so quartiles do fall on real level boundaries, but the channel there is too small to support a claim in either direction. Where the target is *near-ceiling* (breast) there is no headroom for any channel to exploit. We scope all experiments to tabular data, following the attack we extend. We therefore make no claim about high-dimensional or non-axis-aligned regimes.

6.4 Is the threshold privileged to the target?

§6.3 establishes that the discretization threshold is the channel that matters. It does not establish that the threshold is *information the service discloses*. LIME’s default `QuartileDiscretizer` places its bin edges at the 25/50/75 percentiles of the explainer’s background data, a statistic of the data distribution that is independent of the model and of the labels. If an adversary can estimate those percentiles from data it already holds, the discretization accelerates extraction without disclosing anything.

We test this directly on LIME itself. Holding the traversal, the seeds, and the RNG fixed, we vary only the data the discretizer is fit on, across three paired arms: `nothresh` (no bin edge), `tgt` (discretizer fit on the target’s X_{train} , what the service exposes), and `aux` (discretizer fit on the adversary’s own auxiliary data). We report thr_{tgt} and thr_{aux} , each measured against `nothresh`, and the privileged advantage $\text{leak} = \text{tgt} - \text{aux}$.

The discretization helps, but it is not a disclosure. Across all twelve cells in which the threshold channel is significantly positive, spanning five datasets and every target family, the adversary’s own discretizer reproduces the gain (Table 5): on crop, thr_{aux} reaches +41.9pp against thr_{tgt} ’s +40.9pp, with differences averaging -0.7 pp and spanning -3.4 to $+1.1$ pp against threshold effects of up to +40.9pp. Exactly one leak term reaches significance, and it runs the wrong way for the service: on crop/KNN the adversary’s self-computed grid is *better* than the target’s own by 2.5pp ($p<.05$). Elsewhere the sign is inconsistent, the self-computed grid being

Table 5: Is the threshold privileged to the target? Every cell where the threshold channel significantly helps ($p < .05$), ordered by effect size. Fidelity gains over a no-threshold baseline, from the target’s own discretizer (tgt) and from one the adversary fits on its auxiliary partition (aux). leak = $\text{tgt} - \text{aux}$, negative favours the adversary, $*p < .05$. Ten paired splits, and tgt reproduces Table 4.

dataset	model	tgt	aux	leak
crop	LR	+40.9	+41.9	-0.9
crop	DT	+38.0	+36.8	+1.1
crop	NB	+24.9	+24.1	+0.9
crop	RF	+22.7	+22.4	+0.3
pendigits	LR	+18.6	+20.1	-1.5
crop	KNN	+15.2	+17.7	-2.5*
mushroom	KNN	+13.9	+15.8	-1.9
breast	NB	+11.2	+11.9	-0.7
pendigits	DT	+10.1	+13.5	-3.4
pendigits	NB	+7.9	+7.8	+0.2
nursery	NB	+5.0	+4.7	+0.3
nursery	KNN	+1.0	+1.3	-0.3

marginally better in seven of the twelve cells, which is what one expects of two estimates of the same quantity rather than of a degraded substitute. The result covers continuous (crop, pendigits), near-ceiling (breast), and categorical (nursery) targets alike. The threshold channel is therefore a *self-computable* property of the data distribution rather than privileged information about the model. This reframes the finding of §6.3: what LIME’s discretization supplies is a data-adaptive grid of candidate cut points that makes boundary-straddling queries cheap, valuable to an attacker, but not something the service is uniquely able to reveal.

How much auxiliary data this requires. Self-computation is bounded by the quality of the adversary’s distributional estimate, and the threat model we inherit permits a range of auxiliary set sizes [12]. The result above fits the discretizer on the full auxiliary partition, the “comprehensive” end of that range, but the attack does not need it. Sweeping the number of samples *per class* the discretizer is fitted on, everything else held fixed (Appendix C), a *single* sample per class already reaches parity on crop (+40.7 against the target’s +40.9 on LR, seventeen points against a 1,275-row training set), and about five suffice on pendigits, whose sixteen wide-range features and ten classes make the quantiles harder to place. What the adversary needs is a small *representative* sample, and “small” means single digits per class, so nothing here rests on the generous end of the range we inherit.

Why obfuscating the channel is unlikely to defend it. Because the channel localises to LIME’s `discretize_continuous` step, disabling or coarsening it is the natural candidate mitigation. Two caveats prevent us from claiming it as a validated defense. The first is the result above: the grid is reconstructible, so withholding it removes a convenience rather than a capability. Second, preliminary attempts to perturb the bins rather than remove them (more bins, rounded edges) did not reduce extraction. Measuring what a

defender actually gains by withholding or distorting the discretization, against an adversary that adapts by self-supplying it, is left to future work.

6.5 What does the adversary need, and does withholding it help?

The two channel ablations each remove one channel from the base attack. Composing them, and adding back the query synthesis of §5, yields the question this paper exists to answer: what does an extraction adversary actually need, and how much of it does the service supply? We answer it with a *ladder* of four adversaries that differ only in what they are given. All four run the same framework, the same counted budget, the same auxiliary pool, the same per-split seeds, the same surrogate family and the same evaluation set:

- BLIND: base traversal, random features, no grid, no explanation call. A prediction-only adversary, and the floor.
- AUTOLYCUS: base traversal with the target’s explanation, its feature ranking *and* its bin edges. The published attack.
- SELF-GRID: base traversal, random features, bin edges from a discretizer the adversary fits on its own auxiliary sample. No explanation call is made.
- CUTPURSE(ours): SELF-GRID plus Phase 1 diverse generation and Phase 2 boundary search. No explanation call is made.

Two of the contrasts this ladder supports answer different questions and should not be conflated. SELF-GRID versus AUTOLYCUS holds the attack phases and the budget fixed and replaces the explanation entirely, ranking and bin edges alike, with the adversary’s own grid, so it measures what explanation access is worth end to end. This is the privacy comparison; §6.4 isolates the grid alone under tighter control and should be read alongside it. CUTPURSE versus AUTOLYCUS varies the attack as well, so it measures attack strength rather than disclosure. We report both and keep them apart.

Three contrasts carry the argument: CUTPURSE – AUTOLYCUS (does the attack need the explanation endpoint at all?), CUTPURSE – SELF-GRID (what do Phases 1–2 add on top of the same grid?), and SELF-GRID – AUTOLYCUS (does a self-computed grid substitute for the service’s, at equal phases?).

The two explanation-free arms make *zero* calls to `explain_instance`: the bin edges they consume are read directly from the adversary’s own discretizer, which is fit on its auxiliary pool when the explainer is constructed and never touches the target.

The grid matters, but its source does not. Compare BLIND against the two grid-holding base-traversal arms. The grid is the largest single effect in the paper: on crop it moves fidelity from 38.9 to 82.3 (LR) and 40.0 to 77.9 (DT), and it is what separates a usable adversary from a hopeless one. But its *source* does not matter. Across all thirty cells SELF-GRID – AUTOLYCUS averages +0.5pp with significance in both directions (7 positive, 4 negative), and neither feature type shows a meaningful residual (+1.2pp on the continuous datasets, -0.8pp on the categorical ones, against threshold effects of up to +41pp). An adversary that computes its own discretization from an auxiliary sample is, to within noise, exactly as well off as one handed the target’s explanation, and here that arm reads *nothing*

Table 6: The adversary ladder. Fidelity (%) for each adversary, then the three contrasts that carry the argument. SG-AUT is the disclosure comparison: it swaps the target’s explanation for the adversary’s own grid at fixed phases. DEFENDED is the service that returns attribution weights but withholds the bin edges. All arms share the counted budget Q , the auxiliary pool, the per-split seeds and the evaluation set; ten independent target instantiations per cell, paired Wilcoxon, $*p<.05$, $p<.01$. SELF-GRID and CUTPURSE issue zero explanation calls.**

dataset	model	fidelity					contrast		
		BLIND	DEFENDED	AUTOLYCUS	SELF-GRID	CUTPURSE	SG-AUT	CUTPURSE-AUT	CUTPURSE-DEF
crop	LR	38.9	39.5	80.4	82.3	91.6	+1.8	+11.1**	+52.1**
crop	NB	57.5	56.0	80.9	72.7	75.4	-8.2**	-5.5*	+19.5**
crop	KNN	53.6	51.4	66.6	61.4	63.1	-5.3*	-3.5	+11.6**
crop	DT	40.0	39.7	77.6	77.9	89.9	+0.3	+12.3**	+50.3**
crop	RF	64.3	59.5	82.2	86.1	91.3	+3.9*	+9.2**	+31.9**
pendigits	LR	57.2	48.9	67.5	70.3	86.1	+2.7	+18.5**	+37.1**
pendigits	NB	66.7	65.2	73.1	72.4	84.1	-0.8	+11.0**	+18.9**
pendigits	KNN	62.5	62.5	62.8	64.2	68.9	+1.4*	+6.1**	+6.4**
pendigits	DT	30.7	32.6	42.7	48.2	49.6	+5.5	+6.9*	+16.9**
pendigits	RF	47.6	50.5	49.8	56.7	73.4	+6.9**	+23.6**	+22.9**
breast	LR	68.9	72.8	62.6	60.7	78.0	-1.9	+15.4	+5.2
breast	NB	74.1	75.6	86.8	80.7	88.9	-6.1	+2.1	+13.3*
breast	KNN	69.1	71.7	71.4	66.2	73.8	-5.2	+2.4	+2.1
breast	DT	63.8	64.0	68.8	72.2	73.4	+3.4	+4.6	+9.4
breast	RF	85.8	75.5	76.6	85.3	82.5	+8.7**	+5.9	+7.0
adult	LR	81.6	81.3	79.5	85.1	86.4	+5.6**	+6.9	+5.1
adult	NB	87.8	87.6	87.7	89.6	89.9	+2.0**	+2.2	+2.2
adult	KNN	82.8	84.0	83.8	84.2	86.7	+0.4	+3.0*	+2.7**
adult	DT	64.4	68.4	71.0	75.1	71.2	+4.1	+0.2	+2.8
adult	RF	81.0	78.9	79.8	83.4	87.5	+3.6	+7.8**	+8.6**
nursery	LR	98.1	98.7	97.9	98.1	97.9	+0.2	+0.0	-0.8**
nursery	NB	84.0	79.1	84.1	90.4	80.9	+6.3**	-3.2	+1.7
nursery	KNN	69.0	80.5	81.5	76.5	71.0	-5.0**	-10.5**	-9.5**
nursery	DT	87.1	89.9	90.5	88.4	88.7	-2.0	-1.8*	-1.2
nursery	RF	86.2	90.3	90.8	90.7	88.7	-0.1	-2.1*	-1.6*
mushroom	LR	85.3	92.3	83.6	82.2	80.8	-1.4	-2.8	-11.5**
mushroom	NB	71.3	65.8	65.6	76.8	79.7	+11.1	+14.0*	+13.8*
mushroom	KNN	65.5	63.7	77.6	72.6	70.4	-5.1*	-7.2	+6.7
mushroom	DT	72.3	81.7	83.7	78.6	77.3	-5.2	-6.5*	-4.4
mushroom	RF	66.8	81.5	80.7	74.1	75.7	-6.6	-5.0	-5.7*
mean		68.8	69.6	76.3	76.8	80.1	+0.5	+3.8	+10.4

from the explanation, neither the ranking nor the edges, so the substitution is complete rather than partial.

Which phase does the work. A component-level decomposition (Appendix A) separates the two additions. They contribute about equally on average (+1.6pp for boundary search, +1.8pp for diverse generation), but which dominates is target-dependent in the way the weaknesses they address predict: naive Bayes gains almost entirely from diverse generation (W1, +6.4pp), logistic regression and random forest almost entirely from boundary search (W2, +2.5 and +4.7pp).

What the query-synthesis phases add. The contrast CUTPURSE – SELF-GRID isolates Phases 1–2 given an identical grid. It is +3.3pp over all cells, and the aggregate hides a sharp and mechanically predictable split: on the four datasets with continuous features it is +5.9pp with *nine significant gains and no significant losses*, while on the two fully categorical datasets it is -1.7pp. The same split governs the comparison against the published attack: CUTPURSE – AUTOLYCUS is +7.0pp (10 significant gains, 1 loss) on continuous data and -2.5pp on categorical data.

The mechanism explains the boundary rather than excusing it. Phases 2 and 3 place queries by snapping a feature to a bin edge. A purely categorical dataset has no continuous axis to discretize,

so there are no edges to snap to and the two phases degenerate to a random-feature traversal that is strictly worse than following the explanation. This is the same scope condition that made the threshold channel flat on adult in §6.3, arriving from the other direction, and it bounds the attack honestly: *CUTPURSE is an attack on tabular targets with continuous features*. Where that condition holds it dominates the explanation-guided baseline while querying no explanation at all. Where it does not, the explanation retains genuine value, most visibly on nursery/KNN, where AUTOLYCUS reaches 81.5 against 71.0 for CUTPURSE.

Neither part suffices alone: AUTOLYCUS already holds a grid and is still beaten on continuous data, while BLIND runs the same traversal without one and is far worse than either. The gain is an interaction between bisection and grid-snapped coordinates.

Withholding the bin edges does not defend. DEFENDED is the service that returns attribution weights but suppresses the discretization, the mitigation §4 derives from W3 and §6.3 shows is effective against the attack it targets. On the ladder it is simply a rung below AUTOLYCUS: removing the grid moves the published attack down toward BLIND, collapsing crop/LR from 80.4 to 39.5 and crop/DT from 77.6 to 39.7. It does not move CUTPURSE, which supplies its own grid and finishes +16.3pp above the defended service on the four continuous-feature datasets, significant in thirteen of twenty cells with no significant losses, reaching +52.1 and +50.3pp on the two crop cells the defense hurt most.

The defense compounds rather than delays. A single budget cannot distinguish a defense from a speed bump: if withholding the bin edges only slowed Autolycus down, the gap would close as the budget grew. Figure 2 shows the opposite. The cost of the defense to Autolycus widens monotonically with budget, from 19.9 to 38.0pp on crop/DT and 11.0 to 24.9pp on crop/NB, and the pattern repeats on a second dataset: on pendigits/LR the defense is worth nothing at $Q=100$ (−1.4pp) and 18.6pp by $Q=1000$. Bin-edge stepping is the mechanism by which the traversal converts budget into boundary information. Withholding it does not delay that conversion, it removes it. The distance from the defended service to CUTPURSE widens for the same reason, reaching +52.1pp on crop/LR and +50.3pp on crop/DT at the largest budget.

Two features of the figure are worth stating rather than leaving to the reader. At the smallest budget CUTPURSE is often *behind* both baselines (crop/KNN −7.1, pendigits/LR −9.4pp at $Q=100$): Phase 2 spends roughly 80 queries on bisection, which at $Q=100$ is most of the budget, and the curves cross by $Q=250$. And on pendigits/KNN the defense gap is flat at approximately zero across every budget, one cell in which the threshold channel does no work at all.

Against an adversary of this kind the mitigation therefore buys none of what it was meant to buy: it forfeits the transparency the explanation was published for and degrades the explanation for legitimate users, while leaving achievable fidelity where it was. It may still be worth something against an adversary with weaker auxiliary data, a regime we do not evaluate.

On the two purely categorical datasets we report no aggregate gain, and Figure 6 shows the reason rather than merely the outcome. There the defense barely moves Autolycus either, the withheld and undefended curves lying on top of one another at every budget, because the channel being withheld does not exist to withhold. On

nursery CUTPURSE trails both baselines at every budget we test and converges toward them as the budget grows, ending level with Autolycus on LR and still 10.5 points behind on KNN. On mushroom the outcome is target-dependent rather than uniformly negative: naive Bayes gains +14.0 against Autolycus while KNN loses 7.2. Neither the attack nor its mitigation is meaningful where there is no continuous axis to discretize.

7 Discussion and Limitations

Where the leak lives. Our decomposition recasts explanation-guided extraction as a question about *discretization thresholds*. The exploitable signal is not the attribution magnitude (§6.2) nor, as prior framings imply, the boundary *geometry*, but the discretization threshold LIME publishes for free (§6.3). SHAP, which reports magnitude without a threshold, carries no comparable signal *on average*.

It does carry one in a specific and explicable case. Exact Tree-SHAP [7] against a tree target is significantly positive on MUSHROOM (DT +9.8, RF +6.9pp, both $p < .01$), and the mechanism is visible: for a tree the features with the largest attributions are the ones tested at the split nodes, so a ranking computed exactly over that structure points at the coordinates where the decision actually changes. That is the attribution channel approximating a threshold rather than replacing it, and it does not generalise: the same explanation costs 11.2 points on logistic regression for the same dataset, because a linear target has no split nodes for the ranking to recover. The practical consequence is still asymmetric: an MLaaS provider can serve SHAP-style magnitude attributions at substantially lower extraction risk than LIME-style discretized explanations, with tree targets the case to watch.

A second consequence is about reach rather than risk. Because CUTPURSE never calls an explanation endpoint, it is not restricted to services that publish explanations at all: the same attack runs unchanged against an ordinary prediction API, which explanation-guided extraction cannot target. The population of services exposed to this attack is therefore strictly larger than the population exposed to the attack it derives from.

Scope and limitations. The threshold channel is not universal, and we do not claim it is. It is load-bearing only for continuous tabular targets with fidelity headroom (crop, pendigits). On categorical features it is roughly an order of magnitude weaker, at most +5 to +6pp against +40.9pp on crop, since discretizing a few ordinal levels yields few usable cut points, and on near-ceiling targets it is absent because the attack is already saturated.

CUTPURSE inherits the same boundary, and for the same reason: it places queries by snapping a feature to a bin edge, so it operates only where that channel exists. The split reported in §6.5, gains on the four continuous-feature datasets and none on nursery or mushroom, is therefore what a missing channel predicts rather than an unexplained failure.

The categorical case deserves a sharper statement than “the channel is weaker there,” because the bins on those datasets are not merely weak but *spurious*. Following Autolycus, we construct LIME without declaring which columns are categorical, so its discretizer treats every column as continuous and places cuts at the quartiles of an integer label encoding. A “bin edge” on mushroom

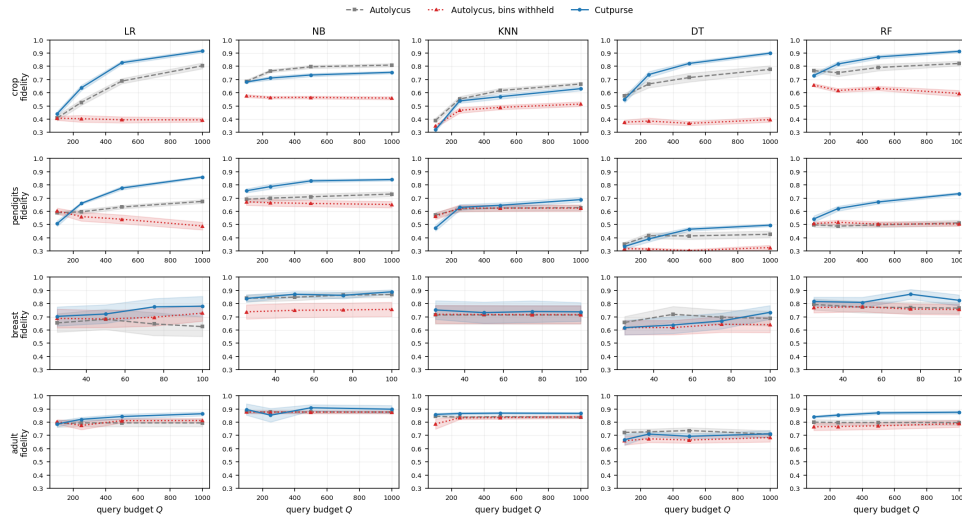


Figure 2: Fidelity vs. query budget for Autolyucus (grey, dashed), Autolyucus with bin edges withheld (red, dotted) and CUTPURSE (blue, solid), per dataset (rows) and target (columns), for the four datasets with continuous features. Band is the standard error over ten independent target instantiations. Same splits, seed sets and attack RNG as Table 6, whose cells are the rightmost point of each curve.

is therefore a quantile of an arbitrary ordering of unordered categories, carrying no information about any decision boundary. This predicts exactly what we observe: on those two datasets the sign of every grid-related effect is inconsistent across targets (on mushroom the same grid is worth +13.8pp to naive Bayes and −11.5pp to logistic regression), which is the signature of an artefact of the encoding rather than a channel. Note that this configuration is the *generous* one: had we declared the categorical columns, LIME’s discretizer would skip them entirely and return equality rules of the form *feature=value*, leaving no bin edge to snap to at all. Our setup therefore grants those datasets a threshold channel they would not otherwise have, and it still does not help. We scope CUTPURSE to targets with genuine continuous features and treat the categorical results as a negative control rather than a competitive comparison.

The consequence for a defender is not symmetric with the consequence for an attacker: the substitutability result of §6.4 holds on *all* six datasets, so withholding explanations does not protect a categorical-feature service either. It merely leaves CUTPURSE without an edge to exploit.

Two measurement caveats bound what our per-cell numbers can support. First, *evaluation-set size* governs resolution: fidelity on BREAST is measured over 85 held-out points, so one point is worth 1.18pp, against 255 for crop (0.39pp) and 1648 for pendigits (0.06pp). Effects of a few points are therefore not resolvable on BREAST, and we rest no claim on it, a caution that applies equally to prior work reporting per-cell fidelity on small splits. Second, the crossing-rate objective in Phase 1 is a proxy for decision-cell coverage.

8 Conclusion

Explanation-guided extraction against interpretable tabular classifiers does not depend on the part of the explanation that is specific to the model. Ablating each channel separately shows the attribution

ranking, in both LIME and SHAP, confers no systematic advantage, being centred on zero and inconsistent in sign. What carries the attack is the discretization threshold LIME exposes as a bin edge, worth up to 41 fidelity points and directionally consistent.

That threshold is a quantile of the background distribution rather than a property of the model, which makes the obvious mitigation self-defeating. A service that withholds the bin edges costs the published attack most of its advantage, but an adversary holding a representative auxiliary sample estimates an equivalent grid and recovers it. We demonstrate this with CUTPURSE, which queries no explanation endpoint at all: against the defended service it gains +16.3 fidelity points on targets with continuous features, against the undefended service +7.0, and the margin over the defended service *widens* with budget rather than closing, because withholding the bin edges removes the mechanism by which the traversal converts queries into boundary information. Because it needs no explanation, the attack also applies to ordinary prediction APIs, which explanation-guided extraction cannot target at all.

The practical reading for a provider is that suppressing explanations is the wrong lever against an adversary that holds auxiliary data of its own. It forfeits the transparency the explanation was published for and leaves that adversary no worse off, because the withheld quantity was never the provider’s to keep. Against an adversary without such data the lever may still do something, and locating that crossover is the experiment we would run next. Where the attack does not apply is where the channel does not exist: on purely categorical targets there are no meaningful cut points, and there CUTPURSE shows no gain. Characterising what a defender can withhold that an adaptive adversary cannot reconstruct remains open.

Ethical Considerations

This work studies an *attack* (model extraction guided by post-hoc explanations) in order to characterise a concrete privacy risk of explainable machine-learning services. All experiments use publicly available benchmark datasets and target models that we train ourselves. No proprietary system, personal data, or live service was queried, and no human subjects were involved. The techniques extend already-published extraction attacks and are evaluated in a controlled setting, so we assess the incremental disclosure risk as low. Our findings are, on balance, risk-*reducing*: we show that the model-specific content of an explanation contributes little to extraction, and that the component which does contribute is one an adversary can reconstruct without the service’s help. This bounds the incremental risk that publishing an explanation creates, rather than expanding it. The work was not submitted to an external ethics board, as it involves no human-subjects or live-system experimentation.

Open Science

We will release our attack implementation and evaluation harness, together with scripts to reproduce every figure and table, upon acceptance. All datasets used are already public.

AI Use

AI-based tools were used to assist with drafting and revising the text (improving flow and correcting phrasing), to help search for related work, and for pre-submission review. The authors have manually verified, and are responsible for, the accuracy, originality, and integrity of all content, including every technical claim, experiment, and citation.

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A Component decomposition: which phase drives the gain?

Our pipeline adds two components to the Autolycus traversal: diverse query generation (Phase 1) and boundary search (Phase 2). We isolate their contributions with three nested arms on the same seed sets, namely Autolycus alone (base), +boundary (Phase 2, no diverse), and +diverse on top, so that boundary = (+boundary) – base and diverse = full – (+boundary). Figure 3 reports the result.

The two components contribute about equally on average (+1.6 and +1.8pp), but *which* one drives the gain is strongly target-dependent: naive Bayes benefits almost entirely from *diverse generation* (+6.4pp), since a generative model needs distribution coverage, whereas random forest and logistic regression benefit almost entirely from *boundary search* (+4.7, +2.5pp), which places labelled points on the decision surface. In every case the gain is query *placement*, not the attribution ranking: consistent with H1, the components determine *where* queries land, and (§6.2) replacing the explanation’s feature ranking with a random one does not systematically change fidelity. One caveat bounds this decomposition: the boundary term as measured here bundles Phase-2 boundary search with the Phase-3 threshold stepping it inherits from Autolycus, so it should be read as the combined contribution of both rather than of bisection alone. §6.3 isolates the threshold component directly. Like the sweeps of Appendix B, this decomposition uses the explanation-reading configuration, so it measures what the two components add *given* the target’s explanation, while §6.5 measures what they add without it.

The split is also strongly *dataset-dependent* (Figure 4): diverse generation carries almost the entire gain on ADULT/NB (+34pp, a generative target starved of coverage), while boundary search dominates on the tree targets of CROP and the linear targets of BREAST. Neither component is tied to a particular dataset, both being query *placement* mechanisms, which is why the aggregate in Figure 3 splits evenly.

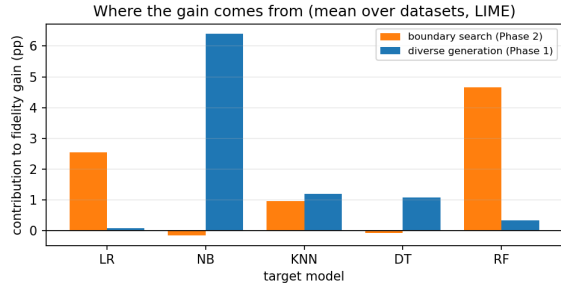


Figure 3: Contribution of each component to the LIME fidelity gain over Autolytus (percentage points, mean over datasets, top budget). Boundary search dominates for RF/LR; diverse generation dominates for NB.

B Attack performance under matched information access

This experiment runs our three phases with the target’s explanation rather than a self-computed grid, so both sides read the same LIME output and the comparison isolates query *placement*. §6.5 evaluates the CUTPURSE attack, which is the one the paper proposes. This appendix and Appendix A predate the multi-split protocol of §6.1 and are reported unchanged on the original single split over five datasets, omitting pendigits, so that they stay comparable to the published baseline.

Figure 5 plots surrogate fidelity against query budget for the explanation-reading configuration and the Autolytus LIME baseline. At a single fixed diverse budget it improves fidelity on 18 of 25 target/dataset combinations (10 significant at $p < .05$, against 2 significant regressions), with the most consistent gains on the non-tree targets that have headroom (LR/NB on crop, nursery, and mushroom).

Part of this margin is an accounting effect worth stating plainly. Autolytus’s traversal terminates when its queue empties, which on several targets happens well before the budget is exhausted. Its curve then flattens because it *cannot* spend the remaining budget, not because additional queries would not help. Our Phase-1 and Phase-2 components keep generating admissible queries, so they continue to convert budget into fidelity. Read at equal *budget* the comparison therefore flatters this configuration, and read at equal *consumed queries* it would flatter Autolytus. We plot budget because that is the baseline’s convention and the quantity a defender actually meters. The $k=3$ control in Appendix D, where the baseline does spend its full budget, isolates how much of the margin is saturation rather than better placement.

The same attack helps under SHAP. Repeating this with SHAP as the explainer ($n=5$, $k=3$) gives the same picture: the same configuration improves over Autolytus-SHAP on the same non-tree targets (LR/NB/KNN, largest where the baseline has headroom, e.g. nursery and mushroom), while tree targets sit near their fidelity ceiling and show no gain (Figure 7, Appendix D). That the same boundary-and-diversity machinery helps under *both* explainers, independent of the particular attribution scores it consumes, is the first indication that the attribution channel is not what drives the

Table 7: Fidelity gain over a no-threshold baseline with the adversary’s discretizer fitted on n samples *per class* rather than on its full auxiliary partition. Cells and tgt column as in Table 5. Ten paired splits.

dataset	model	tgt	aux, fitted on n per class				aux
			$n=1$	$n=2$	$n=3$	$n=5$	
crop	LR	+40.9	+40.7	+43.9	+40.9	+42.8	+41.9
crop	DT	+38.0	+38.2	+38.2	+38.6	+37.3	+36.8
crop	NB	+24.9	+23.4	+23.4	+24.5	+24.1	+24.1
crop	RF	+22.7	+20.5	+20.7	+23.4	+22.1	+22.4
pendigits	LR	+18.6	+12.9	+15.4	+16.5	+16.6	+20.1
crop	KNN	+15.2	+10.9	+16.4	+17.1	+15.9	+17.7
breast	NB	+11.2	+6.0	+10.9	+9.7	+11.9	+11.9
pendigits	DT	+10.1	+9.3	+9.3	+10.9	+10.6	+13.5
pendigits	NB	+7.9	+5.2	+7.8	+7.8	+6.8	+7.8

gain (H1). The LIME-specific advantage of a free *threshold* is isolated separately in §6.3.

Diverse-budget sensitivity. The diverse generation (Phase 1) is regime-dependent: its optimal budget is roughly bimodal, helping monotonically on high-headroom targets (best at the maximum budget, e.g. NB/adult: 0.61→0.96 from no-diverse to full) and hurting on near-ceiling or small continuous targets (best with no diverse). Because the budget is not the contribution, the headline figure fixes it to a single value (cap=15, the best global choice: +3.4pp mean, 18/25 wins) rather than tuning per target. Fixing it this way costs accuracy on the targets that prefer a different budget, and we prefer that cost to a per-target tuning that would not be available to a real attacker.

C How little auxiliary data rebuilds the grid

§6.4 shows that a discretizer the adversary fits on its own auxiliary partition reproduces the target’s. That partition is the “comprehensive” end of the range the threat model inherits [12]. Here we find the other end, sweeping the number of samples *per class* the discretizer is fitted on while holding the attack, the seed set, the budget and the RNG fixed, so that only the estimate of the grid varies (Table 7). We run this on the datasets with continuous features: on the fully categorical ones the bin edges are artefacts of the integer encoding (§7), so how fast that artefact converges is not informative.

The requirement is very small, and crop and pendigits bound it from either side. On crop the adversary is at parity with *one* sample per class: +40.7 against the target’s +40.9 on LR, +38.2 against +38.0 on DT. Only KNN lags at $n=1$, and it catches up by $n=2$. Pendigits converges more slowly, as its shape predicts: sixteen wide-range continuous features to place quartiles on and ten classes rather than seventeen. Its tree and generative targets are nonetheless at parity almost at once (+9.3 against the target’s +10.1 on DT at one sample per class, +7.8 against +7.9 on NB at two), while LR climbs +12.9 → +15.4 → +16.5 → +16.6 and is still some two points short of the target’s +18.6 at five per class, closing only with the full partition. Breast sits between them, at parity by $n=2$.

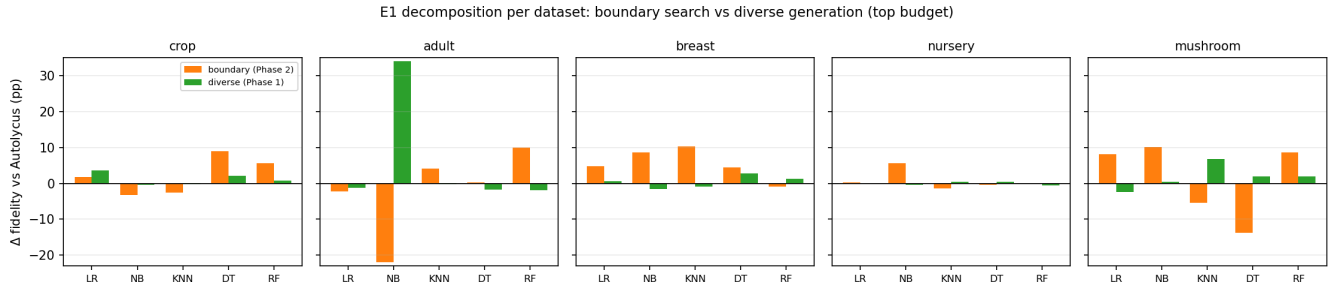


Figure 4: Per dataset: contribution of boundary search (Phase 2, orange) and diverse generation (Phase 1, green) to the LIME fidelity gain over Autolycus (percentage points, top budget, mean over $S=10$ seed sets), one panel per dataset, grouped by target model. Shared y -axis; the ADULT/NB diverse bar (+34pp) sets the scale.

A quartile is a cheap statistic, and a handful of points per class estimates it well enough to recover most of the gain: on crop the columns to the left of aux are, to within noise, the same numbers, and on pendigits they reach 83 to 100 per cent of it. An adversary holding a small fraction of the auxiliary pool is therefore no worse off. Re-estimating the bins online from the accumulated query set is the one variant that does worse, because the attack concentrates queries near decision boundaries and its own history is therefore a biased estimator of the marginal distribution the bins summarise.

D Query-budget sweeps

This appendix collects two kinds of budget sweep. Figure 6 is the categorical companion to Figure 2 and, like it, evaluates CUTPURSE against the defended and undefended service under the multi-split protocol. The remaining figures instead run our three phases with the target’s explanation rather than a self-computed grid, so that

both sides read the same output and the comparison isolates query *placement*. The CUTPURSE attack itself is evaluated in §6.5.

Figure 7 is the SHAP companion to Figure 5, discussed in §B.

Seed count and feature budget. The main experiments use the Autolycus setting of $k=3$, $n=1$. Two alternatives, run under the identical protocol and fixed diverse budget, show that the reported margin is not a seed effect. At $k=1$, $n=3$ the margin over Autolycus is the largest we measure (+4.5pp), but this is an artefact of *base saturation*: exploring a single feature per sample empties the traversal queue after 25.6% of the top budget on average, so the baseline cannot spend the rest, while our diverse generation keeps placing queries. At $k=3$, $n=3$ the baseline spends 88.6% of the budget and the margin *falls* to +1.9pp, against +3.4pp at the headline setting. Additional seeds therefore help Autolycus as much as they help us, and the larger $k=1$ margin reflects the baseline’s budget under-utilisation rather than a seed advantage of our pipeline.

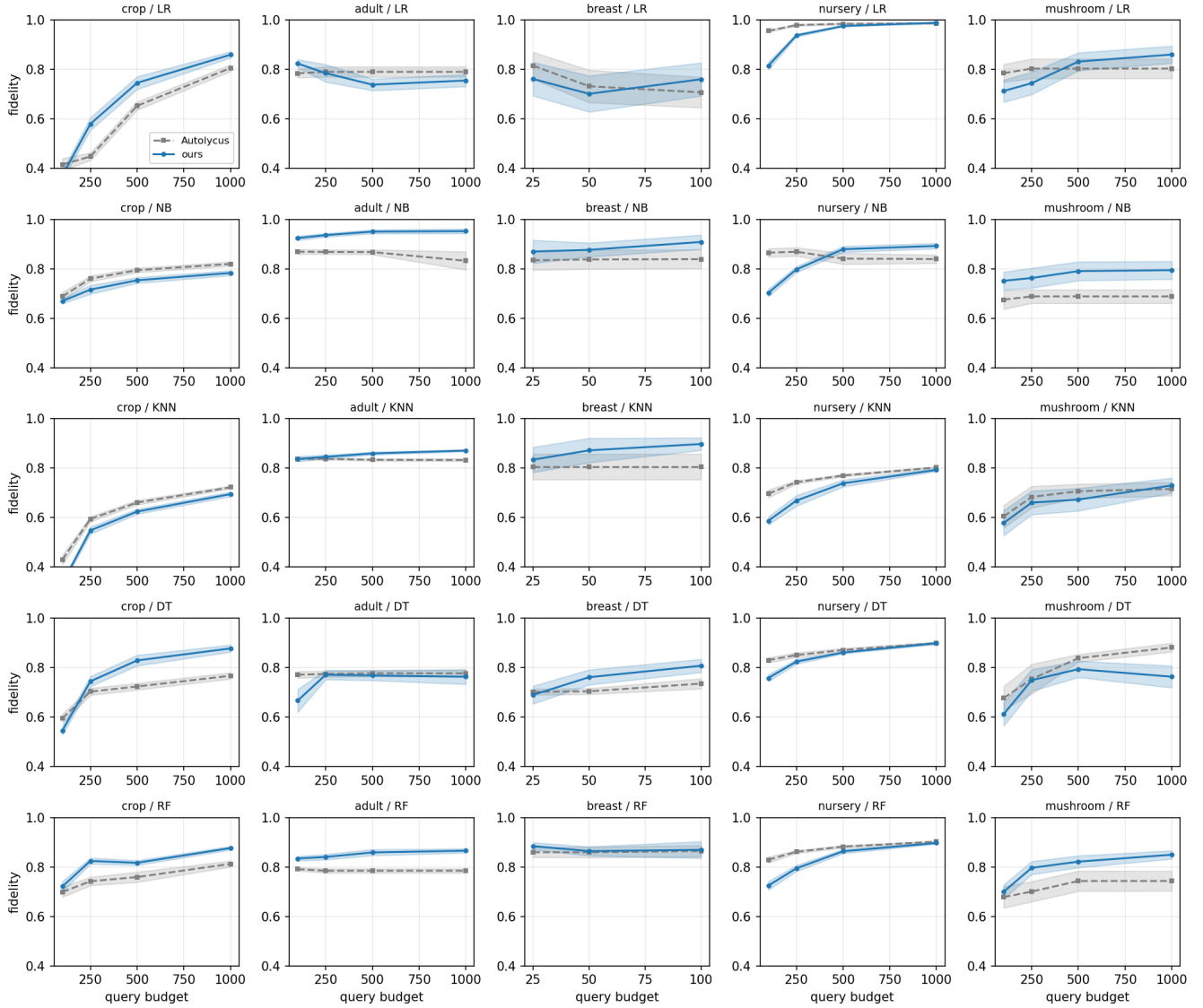


Figure 5: Fidelity vs. query budget, our phases reading the target’s explanation (solid, blue) vs. Autolycus-LIME (dashed, grey), per target (rows) and dataset (columns). Band is the standard error over $S=10$ seed sets; $n=1$, $k=3$, diverse budget fixed at $\text{cap}=15$. Shared y -axis (0.4–1.0).

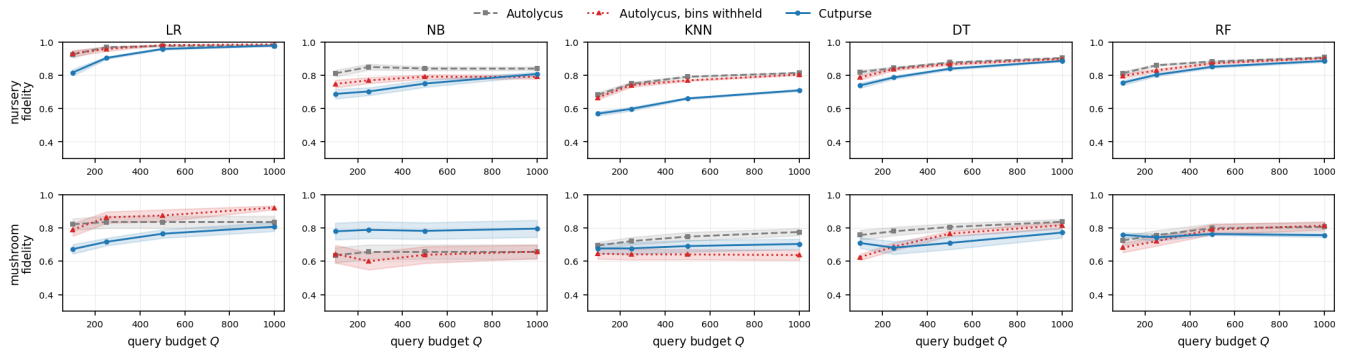


Figure 6: Defense sweep on the two purely categorical datasets, the companion to Figure 2. Withholding the bin edges costs Autolytus almost nothing here, since integer-encoded categories give the discretizer no meaningful cut points, and CUTPURSE has no edge to snap to either.

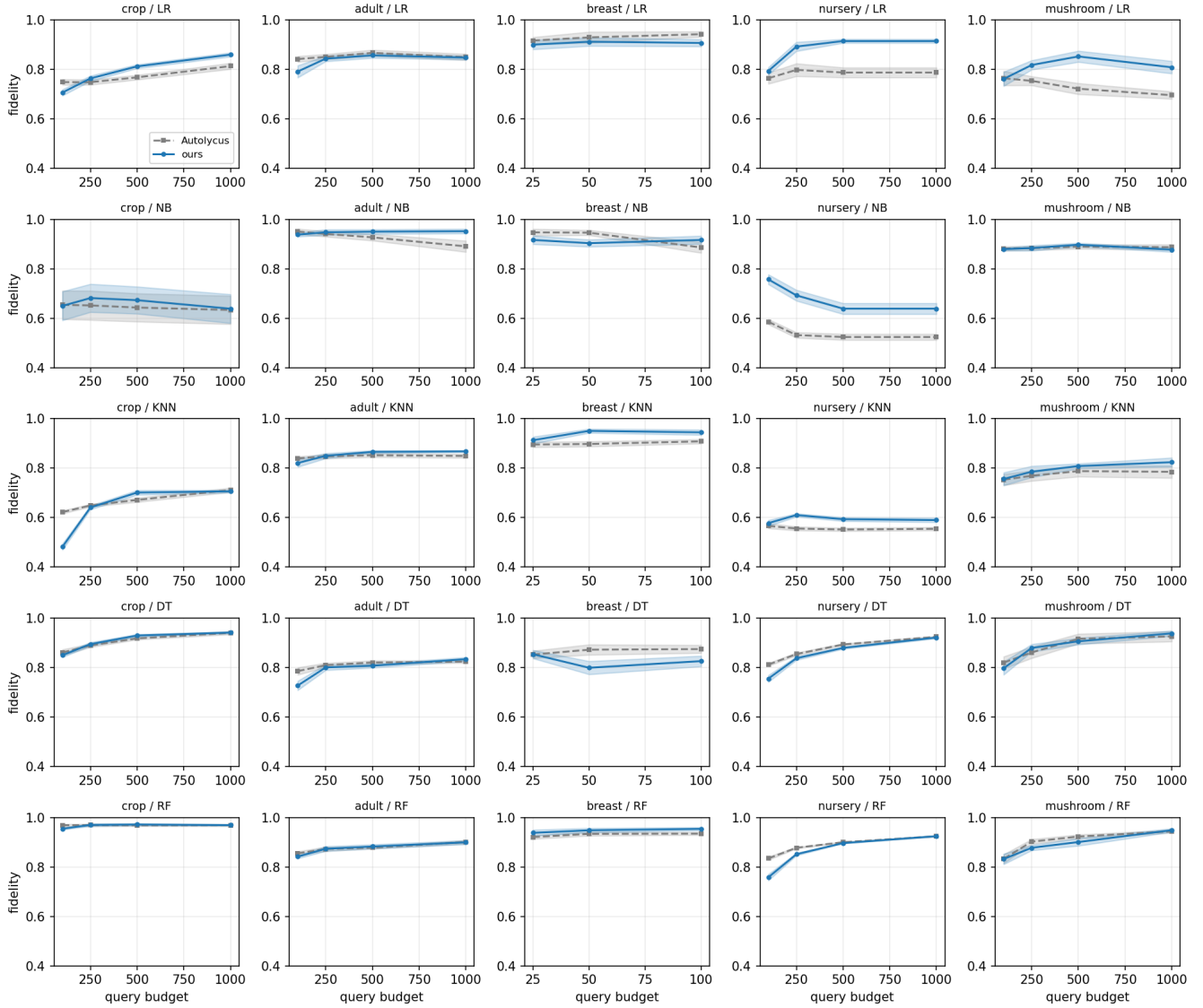


Figure 7: SHAP companion: surrogate fidelity vs. query budget. Our phases reading the target’s SHAP output (solid, blue) vs. Autolytus-SHAP (dashed, grey), per target (rows) and dataset (columns). Shaded band is the standard error over the $S=10$ seed sets; $n=5$, $k=3$. All panels share a fixed y -axis (0.4–1.0) for cross-panel comparability.